

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('C:/Users/india/Desktop/Projects/New
folder/Expanded_data_with_more_features.csv')
df.head()
```

Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType
0	female	NaN	bachelor's degree	standard
1	female	group C	some college	standard
2	female	group B	master's degree	standard
3	male	group A	associate's degree	free/reduced
4	male	group C	some college	standard

ParentMaritalStatus	PracticeSport	IsFirstChild	NrSiblings
married	regularly	yes	3.0
married	sometimes	yes	0.0
single	sometimes	yes	4.0
married	never	no	1.0
married	sometimes	yes	0.0

WklyStudyHours	MathScore	ReadingScore	WritingScore
< 5	71	71	74
5 - 10	69	90	88
< 5	87	93	91
5 - 10	45	56	42
5 - 10	76	78	75

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30641 entries, 0 to 30640
Data columns (total 15 columns):
```

#	Column	Non-Null Count	Dtype
0	Unnamed: 0	30641 non-null	int64

```

1  Gender      30641 non-null object
2  EthnicGroup 28801 non-null object
3  ParentEduc  28796 non-null object
4  LunchType   30641 non-null object
5  TestPrep    28811 non-null object
6  ParentMaritalStatus 29451 non-null object
7  PracticeSport 30010 non-null object
8  IsFirstChild 29737 non-null object
9  NrSiblings  29069 non-null float64
10 TransportMeans 27507 non-null object
11 WklyStudyHours 29686 non-null object
12 MathScore    30641 non-null int64
13 ReadingScore 30641 non-null int64
14 WritingScore 30641 non-null int64
dtypes: float64(1), int64(4), object(10)
memory usage: 3.5+ MB

```

```
df.isnull().sum()
```

```

Unnamed: 0      0
Gender          0
EthnicGroup    1840
ParentEduc     1845
LunchType      0
TestPrep      1830
ParentMaritalStatus 1190
PracticeSport   631
IsFirstChild    904
NrSiblings     1572
TransportMeans  3134
WklyStudyHours  955
MathScore       0
ReadingScore    0
WritingScore    0
dtype: int64

```

#Dropping Column

```
df = df.drop('Unnamed: 0', axis = 1)
df.head()
```

	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	\
0	female	NaN	bachelor's degree	standard	none	
1	female	group C	some college	standard	NaN	
2	female	group B	master's degree	standard	none	
3	male	group A	associate's degree	free/reduced	none	
4	male	group C	some college	standard	none	

```

ParentMaritalStatus PracticeSport IsFirstChild NrSiblings
TransportMeans \

```

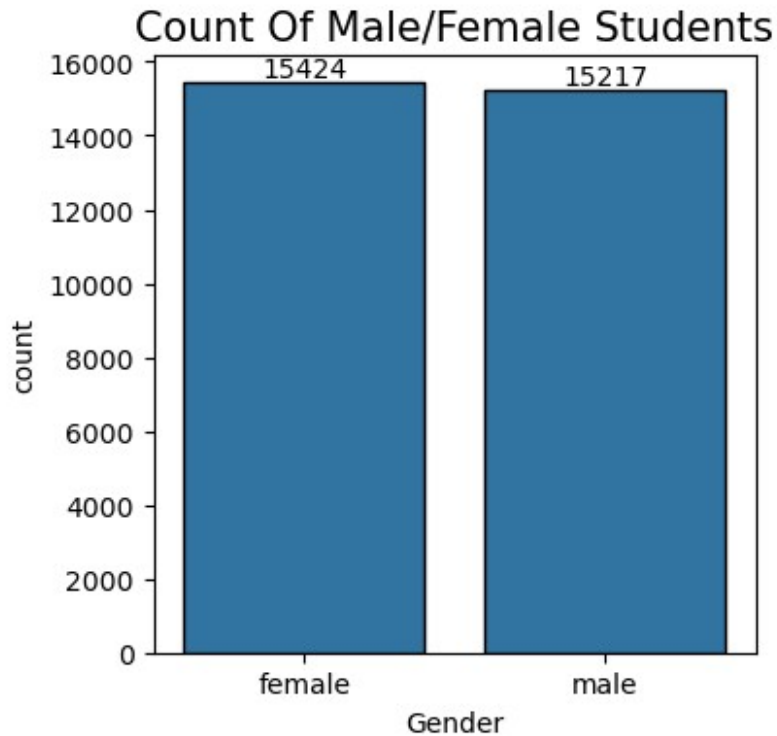
0		married	regularly	yes	3.0
school_bus					
1		married	sometimes	yes	0.0
NaN					
2		single	sometimes	yes	4.0
school_bus					
3		married	never	no	1.0
NaN					
4		married	sometimes	yes	0.0
school_bus					

	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	< 5	71	71	74
1	5 - 10	69	90	88
2	< 5	87	93	91
3	5 - 10	45	56	42
4	5 - 10	76	78	75

```
df.describe()
```

	NrSiblings	MathScore	ReadingScore	WritingScore
count	29069.000000	30641.000000	30641.000000	30641.000000
mean	2.145894	66.558402	69.377533	68.418622
std	1.458242	15.361616	14.758952	15.443525
min	0.000000	0.000000	10.000000	4.000000
25%	1.000000	56.000000	59.000000	58.000000
50%	2.000000	67.000000	70.000000	69.000000
75%	3.000000	78.000000	80.000000	79.000000
max	7.000000	100.000000	100.000000	100.000000

```
plt.figure(figsize = [4,4])
Ar = sns.countplot(x = 'Gender', data = df,edgecolor = 'black')
Ar.bar_label(Ar.containers[0])
plt.title("Count Of Male/Female Students", fontsize = 15)
plt.show()
```

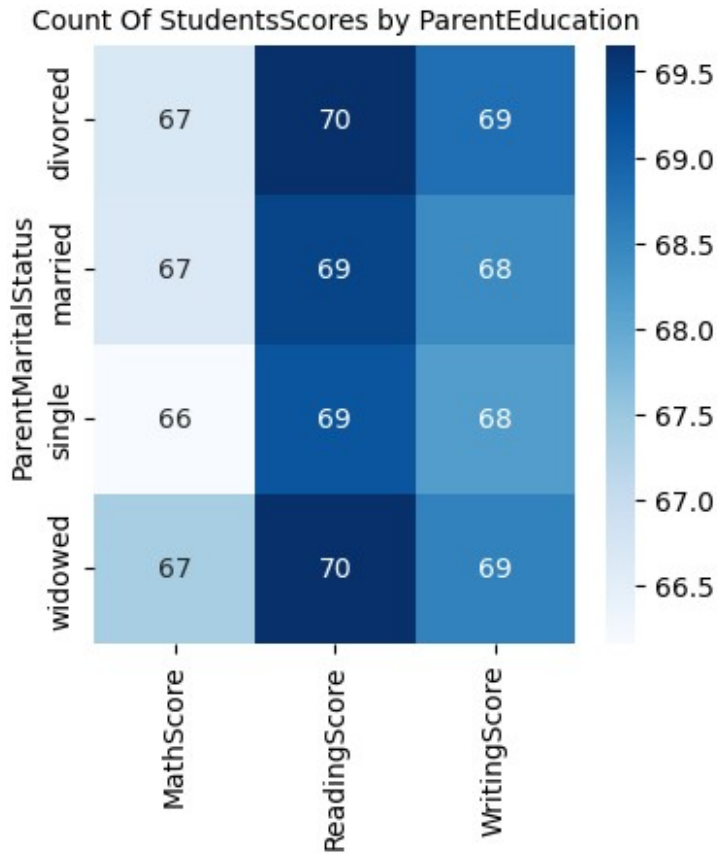


#From the above Chart: We have see that Females are slightly more than Males

```
gr =
df.groupby('ParentEduc').agg({'MathScore':'mean', 'ReadingScore':'mean'
, 'WritingScore':'mean'})
gr
```

ParentEduc	MathScore	ReadingScore	WritingScore
associate's degree	68.365586	71.124324	70.299099
bachelor's degree	70.466627	73.062020	73.331069
high school	64.435731	67.213997	65.421136
master's degree	72.336134	75.832921	76.356896
some college	66.390472	69.179708	68.501432
some high school	62.584013	65.510785	63.632409

```
plt.figure(figsize = [4,4])
sns.heatmap(gr, annot = True, cmap="Blues")
plt.title("Count Of StudentsScores by ParentEducation", fontsize = 10)
plt.show()
```

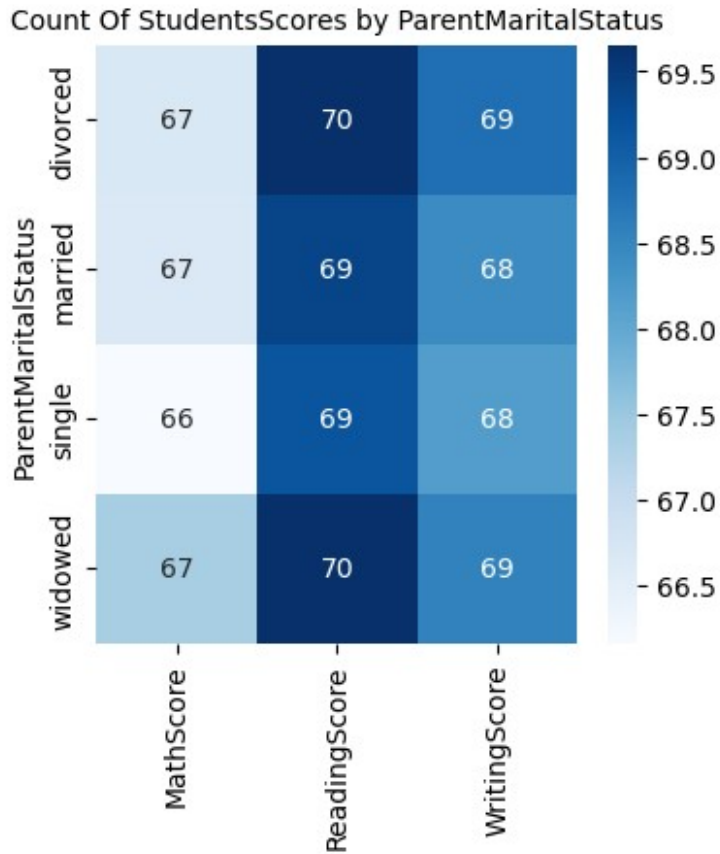


#From the above chart, we conclude that the 'Education of Parents' have a good impact on their childrens scores.

```
grr =
df.groupby('ParentMaritalStatus').agg({'MathScore':'mean', 'ReadingScore':'mean', 'WritingScore':'mean'})
grr
```

ParentMaritalStatus	MathScore	ReadingScore	WritingScore
divorced	66.691197	69.655011	68.799146
married	66.657326	69.389575	68.420981
single	66.165704	69.157250	68.174440
widowed	67.368866	69.651438	68.563452

```
plt.figure(figsize = [4,4])
sns.heatmap(grr, annot = True, cmap="Blues")
plt.title("Count Of StudentsScores by ParentMaritalStatus", fontsize = 10)
plt.show()
```

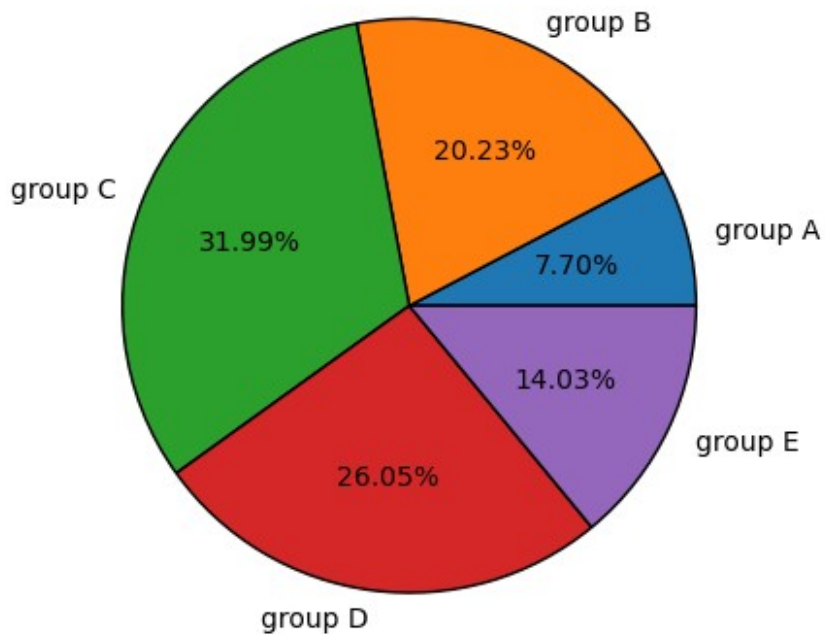


#From the above chart, we have concluded that there is no/negligible impact on Students Score's due to their 'ParentsMaritalStutus'.

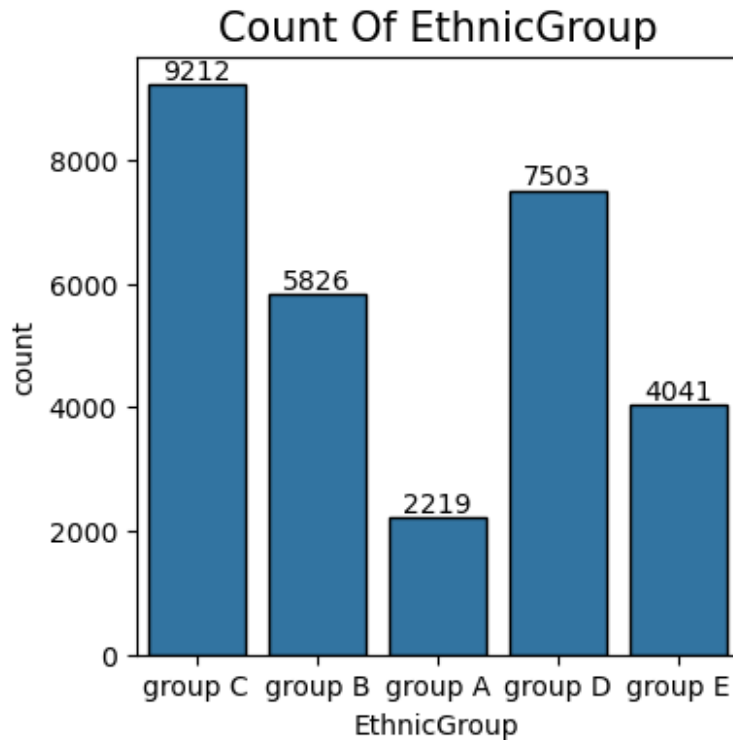
```
GroupA = df.loc[(df['EthnicGroup'] == 'group A')].count()
GroupB = df.loc[(df['EthnicGroup'] == 'group B')].count()
GroupC = df.loc[(df['EthnicGroup'] == 'group C')].count()
GroupD = df.loc[(df['EthnicGroup'] == 'group D')].count()
GroupE = df.loc[(df['EthnicGroup'] == 'group E')].count()
list =
[GroupA['EthnicGroup'],GroupB['EthnicGroup'],GroupC['EthnicGroup'],GroupD['EthnicGroup'],GroupE['EthnicGroup']]
label = ['group A', 'group B', 'group C', 'group D', 'group E']

plt.pie(list, labels = label, autopct = '%1.2f%', wedgeprops={'edgecolor': 'black'})
plt.title("Percentage of Students By EthnicGroup", fontsize = 15)
plt.show()
print(list)
```

Percentage of Students By EthnicGroup



```
[np.int64(2219), np.int64(5826), np.int64(9212), np.int64(7503),  
np.int64(4041)]  
  
plt.figure(figsize = [4,4])  
mg = sns.countplot(x = 'EthnicGroup', data = df, edgecolor = 'black')  
mg.bar_label(mg.containers[0])  
plt.title("Count Of EthnicGroup", fontsize = 15,)  
plt.show()
```



#From the above charts, we conclude that 'Group C' is the highest among the all group and likely 'Group A' is the lowest amount all group.

```
mr =
df.groupby('WklyStudyHours').agg({'MathScore': 'mean', 'ReadingScore': 'mean', 'WritingScore': 'mean'})
mr
```

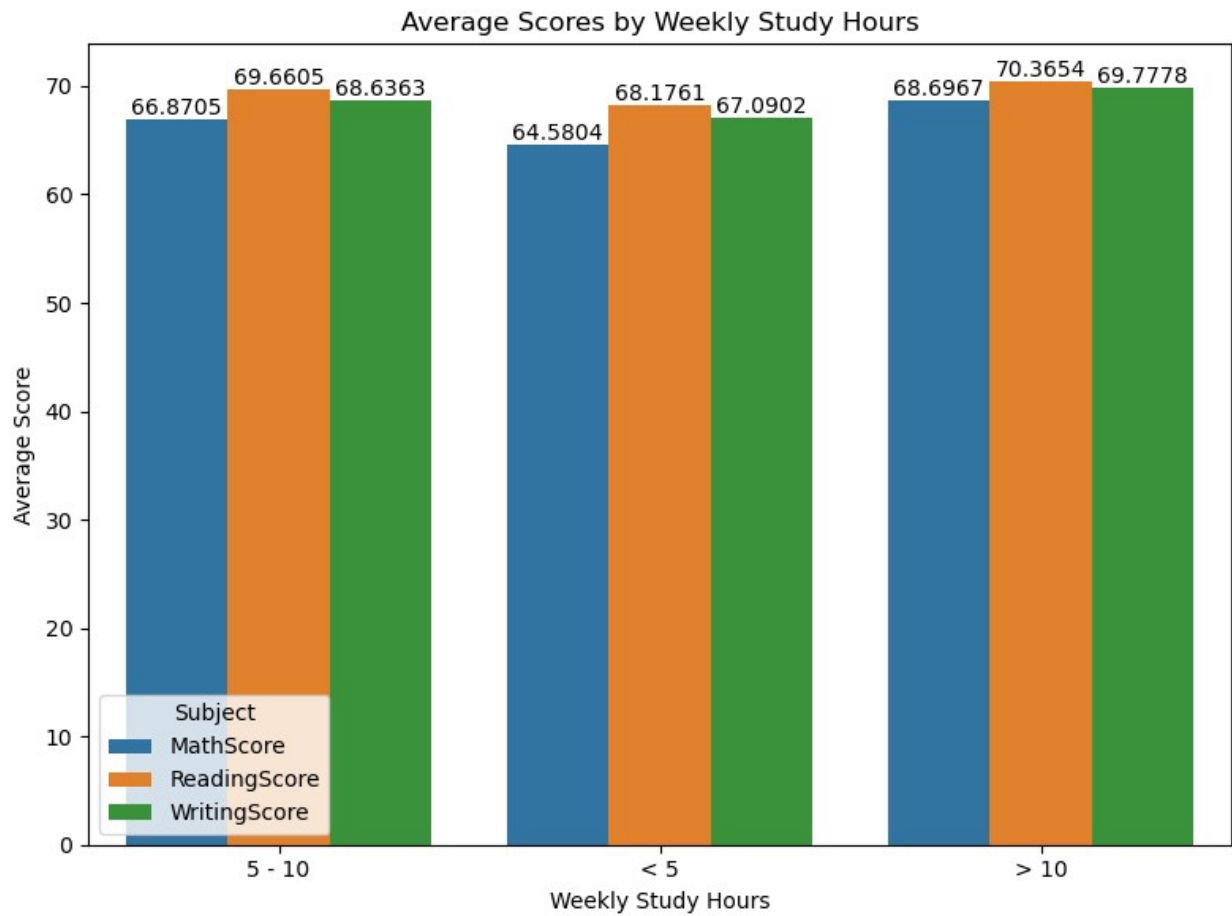
WklyStudyHours	MathScore	ReadingScore	WritingScore
5 - 10	66.870491	69.660532	68.636280
< 5	64.580359	68.176135	67.090192
> 10	68.696655	70.365436	69.777778

```
df_plot = mr.reset_index().melt(id_vars='WklyStudyHours',
var_name='Subject', value_name='AverageScore')

plt.figure(figsize=(8,6))
AT = sns.barplot(x='WklyStudyHours', y='AverageScore', hue='Subject',
data=df_plot, )
AT.bar_label(AT.containers[0])
AT.bar_label(AT.containers[1])
AT.bar_label(AT.containers[2])
plt.title('Average Scores by Weekly Study Hours')
plt.ylabel('Average Score')
plt.xlabel('Weekly Study Hours')
plt.legend(title='Subject')
```



```
plt.tight_layout()
plt.show()
```



#From the above chart, we have concluded that there is no/negligible impact on Students Score's due to their 'WeeklyStudyHours'.